

Instrument Rating (Airplane) Oral Exam Question Bank

A knowledge-focused study resource for the Instrument Rating checkride oral exam. Organized by topic area, with a complete model answer for every question.

Two tips before you start studying: - **Answer the question asked, then stop.** Volunteering extra information beyond what was asked often invites a harder follow-up question that wasn't coming. If the examiner wants more detail, they'll ask for it. - **Speak in your own words, not acronyms.** An examiner can tell when an answer is recited from a mnemonic rather than understood — be ready to explain the substance behind any acronym you use.

1. Certificates, Privileges & Currency

1. What are the eligibility requirements for an instrument rating? **A:** Hold at least a current private pilot certificate (or be concurrently applying for one) with the appropriate category/class rating; be able to read, speak, write, and understand English; pass the instrument knowledge test; and meet the aeronautical experience requirements of 61.65.
2. What's the aeronautical experience breakdown behind the instrument rating? **A:** 50 hours of cross-country PIC time (10 hours in airplanes), and 40 hours of actual or simulated instrument time in the required areas of operation, of which 15 hours must be received from an authorized instructor holding an instrument-airplane rating — including 3 hours of instrument training within 2 calendar months before the practical test, and one cross-country flight under IFR with a filed flight plan covering at least 250 nautical miles along airways or as routed by ATC, with an instrument approach at each airport and three different kinds of approaches using navigation systems.
3. What are the privileges and limitations of an instrument rating? **A:** It allows you to file and fly under IFR, operate in weather below basic VFR minimums (down to the applicable approach minimums), operate in Class A airspace, and — for a commercial pilot — carry passengers for compensation beyond 50 nautical miles or at night. It doesn't remove any category/class/type limitations already on your certificate.
4. What's required to remain current to act as PIC under IFR? **A:** Within the preceding 6 calendar months: 6 instrument approaches, holding procedures and tasks, and intercepting and tracking courses using navigational electronic systems. This can be done in actual or simulated instrument conditions, in the aircraft or an approved simulator/flight training device.
5. You haven't flown any IFR in 7 months. What are your options? **A:** You've lost currency but you're within the grace period — you have another 6 months (through the 12th month) to get current again by flying the 6 approaches/holding/intercept-tracking tasks with a safety pilot, without needing an instructor. If you go beyond 12 months total without currency, you need an Instrument Proficiency Check (IPC) with a CFII to reinstate IFR privileges.

6. Who can act as your safety pilot, and what do they actually need to have? **A:** They need to hold at least a private pilot certificate with the appropriate category and class rating for the aircraft (not necessarily the ability to act as PIC of that specific aircraft — category/class/type is what matters, not full PIC eligibility), a current medical certificate, adequate vision forward and to the sides of the aircraft, and the aircraft must have functioning dual controls.
7. When can you log an instrument approach? **A:** In actual instrument conditions, you can log it as soon as you're in IMC at any point up to and including the final approach segment — you don't have to fly it all the way to minimums if you break out early. In simulated conditions (a view-limiting device), the approach must be flown down to minimums to be logged.
8. What is Part 68 BasicMed, and does it apply to instrument flying? **A:** An alternative to holding an FAA medical certificate for certain flights (aircraft under 6 occupants and 12,500 lbs, no more than 6 passengers, not for compensation or hire). It applies equally to IFR or VFR flight, as long as you stay at or below 18,000 ft MSL and 250 KIAS and meet BasicMed's other requirements.

2. Aircraft Systems & Equipment

9. What additional equipment does 91.205 require for IFR flight, beyond the day/night VFR list? **A:** A two-way radio and navigation equipment suitable for the route being flown, a gyroscopic rate-of-turn indicator (except for aircraft with a third attitude instrument system usable during flight instrument failure), a slip-skid indicator, a sensitive, adjustable altimeter, a clock displaying hours/minutes/seconds, a generator or alternator, and a gyroscopic direction indicator (heading indicator).
10. Your GPS is IFR-approved but doesn't have WAAS. Can you fly it as your only means of navigation under IFR? **A:** No — without WAAS, a standalone IFR GPS can't be your sole means of navigation; you need a way to independently verify position, typically ground-based navigation (VOR) capable of flying the route. WAAS specifically enables filing both the destination and an alternate as GPS-based approaches; without it, either the destination or the alternate must have a non-GPS-based approach available.
11. What is RAIM, and how many satellites does it require? **A:** Receiver Autonomous Integrity Monitoring — a GPS receiver's internal check that the position solution is trustworthy. Determining 3D position requires 4 satellites; a 5th satellite in view is what allows RAIM to actually detect a faulty signal. If you lose RAIM (an LOI — loss of integrity — annunciation), the GPS position can no longer be trusted for navigation.
12. What would you do if you lost RAIM/GPS integrity in flight? **A:** Report it to ATC and request an alternate means of navigation — vectors, or a routing using VORs/airways if you have that capability. If you have no backup navigation source planned, treat it as a reason to get on the ground and re-plan rather than press on; declaring an emergency is reasonable if you're in IMC with no reliable way to navigate.
13. Your VOR/localizer receiver failed its last three 30-day checks, but you only plan to use GPS as primary navigation. Is the aircraft legal for IFR? **A:** Only if the GPS meets IFR navigation requirements on its own (WAAS-capable, or otherwise satisfying 91.205's navigation equipment requirement for the route) and any inoperative VOR equipment has been properly handled under 91.213 — placarded inoperative if not required for the flight, with a mechanic's determination. The VOR check itself (91.171) is only required if you

intend to actually use VOR for IFR navigation.

14. What are the tolerances for a VOR operational check, and what methods are acceptable? **A:** $\pm 4^\circ$ using a VOT (VOR test facility) or a certified ground checkpoint, $\pm 6^\circ$ using a certified airborne checkpoint or a prominent ground landmark, or $\pm 4^\circ$ when comparing two VOR receivers against each other (dual system check). The date, place, bearing error, and your signature must be logged.

15. If your G5 (or similar STC'd glass instrument) is installed, does that mean you no longer need your old mechanical altimeter or airspeed indicator? **A:** Not necessarily — check the specific STC. Many glass-instrument STCs require retaining the original mechanical altimeter, airspeed indicator, and/or turn coordinator because the new unit's pitot-static data isn't certified as primary, only supplementary. The 91.213 decision tree (check the POH/KOEL, then 91.205, then the type certificate data sheet, then any STCs) is what actually determines whether a "broken" old instrument grounds the aircraft — not just whether a modern replacement is present.

16. What anti-icing/de-icing equipment might be found on an IFR-capable airplane, and what should you know about its limitations? **A:** Pitot heat (nearly universal), and depending on the airplane, airframe boots or weeping-wing systems, propeller anti-ice, and heated windshield panels. None of these certify a normal single-engine trainer for flight into known icing — even an aircraft with some anti-ice equipment typically remains prohibited from known icing conditions unless specifically certified for it, and equipment limitations (e.g., boot cycle timing, maximum ice accumulation before effectiveness drops) should be understood before relying on them.

17. What's the purpose of an instrument flight deck check before an IFR flight, and what are you actually checking? **A:** Confirming the aircraft's instruments and IFR-required equipment are functioning correctly and the aircraft is airworthy for instrument flight — attitude indicator erect and stable, altimeter within tolerance of field elevation, DG aligned with the magnetic compass, turn coordinator/ball responding correctly to taxi turns, and navigation equipment tuned and identified.

3. Weather

18. How can you obtain a weather briefing, and what types are there? **A:** 1800wxbrief.com, the ForeFlight/Garmin Pilot app, AviationWeather.gov, or Flight Service by phone or radio. A Standard briefing is complete, used for the first briefing of the day or after 6+ hours since the last one. An Abbreviated briefing updates a previous one. An Outlook briefing is for planning when departure is 6+ hours away, and must be followed by a standard or abbreviated briefing closer to departure.

19. What are AIRMETs and SIGMETs, and how long is each valid? **A:** SIGMETs cover hazards significant to all aircraft — severe icing, severe/extreme turbulence, dust/sandstorms below 3 SM visibility, volcanic ash — valid up to 4 hours (6 for hurricanes). Convective SIGMETs cover severe thunderstorm activity, valid 2 hours. AIRMETs cover less severe conditions of particular concern to lighter aircraft, issued every 6 hours: Sierra (IFR/mountain obscuration), Tango (moderate turbulence, sustained winds ≥ 30 kt, low-level wind shear), Zulu (moderate icing and freezing levels).

20. What types of structural icing are there, and what conditions does icing require? **A:** Rime ice — rough, milky, forms quickly from small supercooled droplets, typical in stratiform clouds. Clear ice — smooth, hard, transparent, the most dangerous and hardest to remove, from larger supercooled droplets that spread before freezing. Mixed ice — a combination of both. Icing requires visible moisture and a temperature at or below freezing at the airframe surface.

21. What is a false glideslope, how do you avoid it, and how would you recognize one? **A:** A false glideslope indication can occur at certain angles well above the true glideslope, often around double the actual glidepath angle. It's avoided by intercepting the glideslope from below at the published glideslope intercept altitude, not from above — if ATC vectors you above the normal intercept altitude, be cautious about "catching" the needle before verifying your position against the final approach fix. A telltale sign is an unusually steep required descent rate to stay on the needle.

22. What's the difference between a localizer's and a VOR's service volume, and why does it matter? **A:** A localizer's usable service volume is roughly 18 nautical miles from the antenna and narrows as you get farther away — about 10° wide at 18 nm, widening to roughly 35° at 10 nm. Outside that volume, a "live" localizer needle isn't reliable, even if it appears to give a sensible indication — a common trap when you're well off course at long range and the needle looks plausible.

23. Flying from an area of very cold, high-pressure air to an area of hot, low-pressure air without resetting your altimeter — are you too high or too low, and what's the risk? **A:** You'd actually be lower than your altimeter indicates when flying toward lower pressure/warmer temperature without resetting — "high to low, look out below." (Conversely, flying from low pressure to high pressure without an update, the altimeter under-reads and you're higher than indicated.) On an approach, the practical risk is descending below a safe altitude or minimums while the altimeter still shows you comfortably above them.

4. Cross-Country & IFR Flight Planning

24. What does an IFR flight plan require, and how do you file, activate, and close one? **A:** Aircraft identification and type/equipment suffix, true airspeed, departure point, requested route and altitude, destination, and estimated time en route plus fuel on board, among the standard elements. File through Flight Service, ATC (a "pop-up" clearance, workload permitting), or a commercial service like ForeFlight — ideally at least 30 minutes before departure. An IFR flight plan in controlled airspace is automatically opened when you receive your clearance and automatically closed by ATC/tower on landing at a towered field; at a nontowered field, you must close it yourself with Flight Service or by radio.

25. What are the elements of a standard ATC clearance, and what's the mnemonic? **A:** CRAFT — Clearance limit, Route, Altitude, Frequency (departure), Transponder code. Example: "Cleared to [destination]. Radar vectors to [fix], then as filed. Maintain 2,000, expect 7,000 ten minutes after departure. Departure frequency 125.65. Squawk 4211."

26. What's an abbreviated departure clearance, and what does it still include? **A:** Used when your filed flight plan is approved with little or no revision — it still includes the clearance limit, any assigned departure procedure and transition, altitude, frequency, and transponder code, just abbreviated in phrasing (e.g., "cleared as filed").

27. What's the difference between a release time, a clearance void time, and a notify time for an IFR departure from a nontowered airport? **A:** The release time is the earliest you may depart. The clearance void time is the latest you may depart — if you haven't departed by then, the clearance is void and you must contact ATC. The notify time is the latest you should advise ATC if you haven't departed, if a separate notify time was given.
28. What's the difference between a SID and an ODP, and how do you tell them apart on a chart? **A:** An ODP (Obstacle Departure Procedure) exists purely for obstacle clearance and is designed to be flown without ATC involvement if not otherwise cleared; it's noted with "(OBSTACLE)" in the takeoff minimums section. A SID (Standard Instrument Departure) also provides obstacle clearance but is primarily designed to simplify clearance delivery and reduce ATC/pilot workload on a busy departure — it's assigned by ATC via a clearance.
29. What is a nonstandard takeoff minimum and climb gradient, and what determines whether you can legally use a given SID/ODP? **A:** Under Part 91, no takeoff minimums are legally required unless the departure procedure specifically mandates them, but a published departure procedure may require a minimum climb gradient (e.g., 260 ft/NM) to a specified altitude to guarantee obstacle clearance. Before accepting that departure, you need to know your airplane can actually achieve that climb gradient at the current weight, temperature, and wind — a climb gradient is a distance-based requirement, not directly the same as a feet-per-minute rate of climb, so it has to be converted using your expected groundspeed.
30. What do you check before departing on any IFR flight, per 91.103? **A:** NOTAMs, weather reports and forecasts, known ATC delays, runway lengths at airports of intended use, alternatives if the flight can't be completed as planned, and fuel requirements — takeoff and landing performance data for any flight.
31. What fuel is required to legally depart IFR? **A:** Enough to fly to the first point of intended landing, then to the alternate if one is required, then to fly for 45 minutes at normal cruise after that.
32. When is an alternate airport required, and what are the weather minimums for filing one (the "1-2-3" rule)? **A:** An alternate isn't required if, for at least one hour before through one hour after your estimated time of arrival, the forecast ceiling and visibility at the destination are at least 2,000 feet and 3 statute miles (for an airport with a published instrument approach). If the destination doesn't meet that, you must file an alternate that meets its own minimums: 600-2 (ceiling/visibility) if it has a precision approach, or 800-2 if only non-precision approaches are available — or the airport's own published alternate minimums if different.

5. ATC Clearances, Lost Communications & Holding

33. If you don't fully understand an ATC clearance, or believe it might not be intended for you, what should you do? **A:** Ask for clarification or a full readback confirmation before complying — don't guess. A clearance inconsistent with your aircraft's performance or navigation capability, or one that might be intended for another aircraft with a similar call sign, is exactly the kind of thing worth double-checking rather than assuming is correct.
34. What is PIC emergency authority? **A:** In an in-flight emergency requiring immediate action, the pilot in command may deviate from any rule of Part 91 to the extent required to meet that emergency, though a

written report may be requested by ATC afterward.

35. What altitude do you fly, and what route, if you lose two-way radio communication under IFR? **A:** Route: fly the route you were **A**ssigned in the last clearance; if being radar **V**ectored, fly direct to the fix/route/airway specified in the vector clearance; if none of those, fly the route you were told to **E**xpect in a further clearance; otherwise, fly the route you **F**iled (AVEF). Altitude: fly the highest of the altitude you were **A**ssigned, the altitude you were told to **E**xpect, or the **M**inimum en route altitude for the route segment (AIM's altitude mnemonic, sometimes remembered alongside AVEF).

36. If your clearance limit is a fix rather than the destination and you haven't received further clearance, what do you do at that fix? **A:** If you've received an Expect Further Clearance (EFC) time, hold at that fix until the EFC time, then proceed on your filed route. Only leave the clearance limit fix to begin an approach at your ETA (or EFC time, whichever is later) if the clearance limit is a fix from which an approach begins — otherwise, hold and wait for the EFC time before proceeding to a fix where an approach can be commenced.

37. What are the standard holding pattern parameters unless otherwise instructed? **A:** Right-hand turns, one-minute legs (1.5 minutes above 14,000 ft). Begin timing the outbound leg when abeam the fix (or when the needle flips from TO to FROM), and adjust the outbound leg length as needed so the inbound leg is the correct duration.

38. Describe the three holding pattern entry procedures. **A:** Direct entry — cross the fix and turn directly into the holding pattern. Parallel entry — cross the fix, turn to parallel the inbound course outbound, then after about a minute turn through roughly 210° to intercept the inbound course. Teardrop entry — cross the fix, turn to a heading about 30° offset from the inbound course into the pattern, fly outbound about a minute, then turn about 210° to intercept the inbound course. Any entry that keeps you within the protected area is acceptable — there's no requirement to use the "textbook" entry for your approach angle.

39. What should you do with your wind correction angle when transitioning from the inbound to the outbound leg of a hold? **A:** Roughly triple the wind correction angle (in the opposite direction) on the outbound leg compared to what's needed to hold the inbound course — since you're only trying to correct for one leg's worth of drift on the way out to set up a clean inbound leg.

40. When must you comply with a charted procedure turn, hold-in-lieu-of-procedure-turn, or teardrop entry on an approach, and when can you skip it? **A:** It must be flown unless: you're given a straight-in approach clearance, you're being vectored to the final approach course by ATC, the approach is via a DME arc, "NoPT" is published for your route of arrival, or you're making a timed approach from a holding fix.

6. Navigation Systems

41. How does VOR navigation work? **A:** A VOR station transmits a reference phase signal that's the same in all directions and a second, variable-phase signal whose phase relative to the reference changes depending on the radial you're on — the receiver compares the two signals' phase difference to determine your bearing from the station.

42. How does a localizer differ from a VOR, and why is it more accurate? **A:** A localizer transmits two overlapping signals at different modulation frequencies (90 Hz and 150 Hz) rather than a phase-comparison signal; on the extended runway centerline, equal amounts of each frequency are received, and being left or right of center shifts the balance toward one frequency or the other. This gives a much narrower, more precise course than a VOR's phase-comparison method — roughly a 700-foot-wide course at the runway threshold.
43. How does a glideslope work? **A:** The same frequency-overlap principle as a localizer, but arranged in the vertical plane from a separate antenna near the runway touchdown zone, giving vertical guidance down to the runway.
44. What's the difference between LNAV, LP, LPV, and LNAV/VNAV approaches? **A:** LNAV provides lateral (course) guidance only, using GPS, with a relatively wide final-segment CDI sensitivity (about 0.3 NM full-scale). LP (Localizer Performance) provides lateral guidance with localizer-like sensitivity (roughly 3–6° full-scale, narrowing as you approach the runway) but no vertical guidance. LNAV/VNAV and LPV both add vertical guidance — LPV specifically uses WAAS to provide ILS-like precision approach guidance down to a Decision Altitude, while LNAV/VNAV uses barometric or WAAS vertical guidance to a DA with typically higher minimums than LPV.
45. What is RNAV, broadly, and what are the different underlying technologies? **A:** Area navigation — the ability to fly a direct or defined path that isn't limited to flying directly to/from a single ground-based navaid. It can be implemented via GPS (most common in GA), VOR/DME, DME/DME, or inertial reference systems (still used in some airline/transport aircraft as a backup).

7. Instrument Approach Procedures

46. What's the difference between a precision and a non-precision approach? **A:** A precision approach (ILS, or GPS-based LPV) provides both lateral and vertical guidance down to a Decision Altitude (DA) — you either continue visually from the DA or execute a missed approach immediately, without leveling off at the DA. A non-precision approach provides lateral guidance only (or vertical guidance not accurate enough to be "precision"), descending to a Minimum Descent Altitude (MDA) which you may level off at and fly until reaching the missed approach point.
47. What is a Visual Descent Point (VDP), and how is it depicted? **A:** A point on a non-precision final approach, depicted as a black triangle on the approach chart's profile view, marking where a normal descent from the MDA to the runway can begin once the required visual references are in sight — descending before the VDP risks an excessively steep visual segment.
48. What conditions must be met to descend below the MDA or DA? **A:** The aircraft must be in a position to make a normal descent to landing using normal maneuvers; the flight visibility must not be below the minimum specified for the approach; and for Category I approaches, at least one of the specified visual references for the runway must be distinctly visible and identifiable (runway threshold, runway markings, runway lights, approach lighting system, threshold lights/markings, touchdown zone lights/markings, or a visual glideslope indicator). If the approach lighting system is the only reference in sight, descent may only continue to 100 feet above touchdown zone elevation unless the red terminating bars or red side-row bars

(or another qualifying reference) also become visible.

49. What are aircraft approach categories, and what determines which one you fly? **A:** Categories A through E (and beyond) are based on the aircraft's approach speed — Category A is under 91 knots, B is 91–120 knots, C is 121–140, D is 141–165. Your category (and thus your minimums) is based on 1.3 times your stall speed in the landing configuration at maximum certificated landing weight, not simply the approach speed you choose to fly, and you may not use a category slower than the one your aircraft's actual speed requires.

50. What is a circling approach, and what's the protected maneuvering area based on? **A:** A circling approach descends to a circling MDA, and once the runway environment is in sight, the aircraft maneuvers visually to land on a runway not aligned with the approach course, remaining at or above the circling MDA until a normal descent to land can be made. The protected area is an arc-defined radius from each runway threshold, sized according to the aircraft's approach category (and whether standard or expanded circling minimums are in use) — losing sight of the airport or leaving that protected area requires an immediate missed approach.

51. What must you do immediately upon reaching a missed approach point (or DA, for a precision approach) without the required visual references? **A:** Execute the missed approach procedure promptly — for a precision approach, this means immediately, without leveling off at the DA first. Apply appropriate power, establish the correct pitch attitude, configure the aircraft per the manufacturer's procedure, and follow the published (or ATC-assigned alternate) missed approach procedure, advising ATC if unable to comply with any part of it.

52. What is a Terminal Arrival Area (TAA), and what's its purpose? **A:** A "T"-shaped design used mostly on RNAV approaches, with two Initial Approach Fixes and a dual-purpose Intermediate Fix/IAF that includes a hold-in-lieu-of-procedure-turn, feeding into the final approach fix. It provides a transition from en route to the approach with minimal ATC/pilot interaction and its own published minimum altitudes (replacing the usual minimum safe altitude circle), typically allowing clearance much farther from the airport than a conventional approach.

53. What's the difference between a visual approach and a contact approach? **A:** A visual approach is ATC-authorized once the pilot has the airport (or the aircraft to follow) in sight, requires a ceiling of at least 1,000 ft and 3 SM visibility at the destination, and clears the pilot to proceed visually clear of clouds — normal VFR cloud clearances don't apply since it's still an IFR procedure. A contact approach requires only 1 SM flight visibility and clear-of-clouds operation, must be requested by the pilot (ATC can't initiate it), and requires the destination to have a published instrument approach — it lets the pilot proceed to land without flying the full published procedure.

8. Flight Instruments & Instrument Errors

54. How does the pitot-static system work, and what happens to the altimeter and airspeed indicator if the static port is blocked? **A:** The pitot tube senses ram air pressure; the static port senses ambient (still) air pressure; the airspeed indicator compares the two, while the altimeter and VSI use static pressure alone against a calibrated reference. If the static port is blocked, the altimeter and VSI will freeze at the

pressure/altitude that existed when the blockage occurred, regardless of actual altitude change, and the airspeed indicator will become inaccurate — reading progressively lower than actual as you climb (since ram air pressure drops with altitude but the trapped static reference doesn't fall with it), and progressively higher than actual as you descend.

55. What are the gyroscopic instruments, and what two properties of gyroscopes do they rely on? **A:** The attitude indicator, heading indicator (directional gyro), and turn coordinator. They rely on rigidity in space (a spinning gyro resists changes to the plane of its rotation) and precession (a force applied to a spinning gyro produces a resulting force 90° later in the direction of rotation).

56. What magnetic compass errors should you know, and how do you remember them? **A:** Variation is the fixed difference between true and magnetic north at a location ("east is least, west is best" — subtract east variation, add west variation, going from true to magnetic). Deviation is compass-card error from the aircraft's own electrical/magnetic fields, unique to each airplane. Dip errors occur during maneuvering: on northerly headings, the compass initially shows a turn in the opposite direction during a turn and lags; on southerly headings it leads ("UNOS" — undershoot north, overshoot south, referring to when to stop your turn relative to the desired heading); during acceleration/deceleration on east/west headings, the compass shows a false turn indication ("ANDS" — accelerate north, decelerate south).

57. What's the primary/supporting instrument method of cross-check, and give an example for straight-and-level flight. **A:** Each flight parameter (pitch, bank, power) has one "primary" instrument that gives the most direct, immediate indication of that parameter in a given maneuver, with other instruments "supporting" that reading. In straight-and-level flight: pitch primary is the altimeter (supported by attitude indicator and VSI); bank primary is the heading indicator (supported by attitude indicator and turn coordinator); power primary is airspeed (supported by tachometer/manifold pressure).

58. What are the components of a glass cockpit — AHRS, ADC, PFD, MFD, and Flight Director — and what does each replace? **A:** The AHRS (Attitude and Heading Reference System) provides attitude and heading data, replacing the mechanical attitude indicator, directional gyro, and turn coordinator. The ADC (Air Data Computer), fed by the pitot-static system, replaces the airspeed indicator, altimeter, and VSI. The PFD (Primary Flight Display) graphically presents this "big six" information on one screen; the MFD (Multi-Function Display) shows moving maps, weather, traffic, and system data, and can typically revert to display PFD information if the primary PFD fails. A Flight Director computes and displays a desired flight path/pitch-bank cue to assist hand-flying or drive the autopilot.

9. Spatial Disorientation & Unusual Attitudes

59. What is spatial disorientation, and why is it especially dangerous under instrument conditions? **A:** A false perception of the aircraft's attitude, position, or motion, caused by relying on the body's vestibular and proprioceptive senses rather than the instruments — those senses are unreliable in flight, especially without outside visual reference (IMC or night). The remedy is trusting and cross-checking the flight instruments rather than physical sensation, and maintaining a disciplined instrument scan.

60. How do you recognize and recover from a nose-low unusual attitude by reference to instruments? **A:** Recognize increasing airspeed and a nose-low, often increasing-bank attitude indication. Reduce power,

level the wings using coordinated aileron and rudder, then smoothly raise the nose to level flight — correcting bank and pitch essentially together rather than one fully before the other.

61. How do you recognize and recover from a nose-high unusual attitude by reference to instruments? **A:** Recognize decreasing airspeed (and possibly an approaching stall warning) with a nose-high attitude. Apply full power, level the wings, and smoothly lower the nose to return to level flight.

62. What should you do if you inadvertently enter IMC while not instrument-rated, or otherwise experience unintended IMC? **A:** Transition immediately and fully to the instruments, hold wings level and a constant airspeed, avoid abrupt control inputs, and either turn back toward known VMC (a standard-rate 180°) or, if instrument-rated and IFR-capable, contact ATC and proceed accordingly. Avoid the temptation to look outside for visual references once committed to the instruments — that's exactly what triggers spatial disorientation in this scenario.

10. Emergency Operations

63. What are the general procedures for lost communications, beyond route and altitude? **A:** First, try to reestablish communication — check the frequency, volume, headset connection, and try a different frequency or a nearby facility. Squawk 7600. If unable to reestablish contact, proceed via the AVEF route and MEA-based altitude logic, and begin your approach at your filed/estimated time of arrival (or as close to it as operationally possible) if no other guidance is available.

64. If your primary flight instruments (attitude indicator, etc.) fail or become unreliable during an approach, what's your plan? **A:** Recognize the failure or inaccurate indication, advise ATC, and rely on your secondary/backup instruments and cross-check to continue safe control of the aircraft — most modern trainers retain a standby attitude source or can fly a non-precision approach using the remaining reliable instruments (airspeed, altimeter, heading, turn coordinator) even without the primary attitude display.

65. What is your responsibility, and ATC's role, if you declare an emergency? **A:** As PIC, you have the authority to deviate from any regulation to the extent required to meet the emergency, though you may be asked to file a report afterward. ATC can provide priority handling and vectors — declaring early, rather than waiting until the situation is critical, gives everyone more room to help.

11. Postflight Procedures

66. What must be documented after an IFR flight? **A:** Any in-flight or postflight discrepancies noted with the aircraft's instruments or equipment, servicing requirements, and flight time/approaches logged for currency purposes. The PIC remains responsible for ensuring any discrepancy affecting airworthiness is addressed before the aircraft's next flight.

12. Full Mock Oral Script — Ask in This Order

Read this section straight through, question by question, the way an examiner runs the actual oral. It pulls every question above into one continuous script with its answer, reordered so systems come first — if you don't know your aircraft, nothing after that matters. Ask the question, let the student answer in full before you check it against the model answer, and only give follow-up prompts if they stall.

Before we touch the airplane: have you read the ACS for this rating? Walk me through the three possible outcomes of this practical test, and what has to happen for each one.

Aircraft systems and equipment. Before we get into airspace or weather, I want to make sure you know this airplane cold.

9. What additional equipment does 91.205 require for IFR flight, beyond the day/night VFR list? **A:** A two-way radio and navigation equipment suitable for the route being flown, a gyroscopic rate-of-turn indicator (except for aircraft with a third attitude instrument system usable during flight instrument failure), a slip-skid indicator, a sensitive, adjustable altimeter, a clock displaying hours/minutes/seconds, a generator or alternator, and a gyroscopic direction indicator (heading indicator).
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11. What is RAIM, and how many satellites does it require? **A:** Receiver Autonomous Integrity Monitoring — a GPS receiver's internal check that the position solution is trustworthy. Determining 3D position requires 4 satellites; a 5th satellite in view is what allows RAIM to actually detect a faulty signal. If you lose RAIM (an LOI — loss of integrity — annunciation), the GPS position can no longer be trusted for navigation.
12. What would you do if you lost RAIM/GPS integrity in flight? **A:** Report it to ATC and request an alternate means of navigation — vectors, or a routing using VORs/airways if you have that capability. If you have no backup navigation source planned, treat it as a reason to get on the ground and re-plan rather than press on; declaring an emergency is reasonable if you're in IMC with no reliable way to navigate.
13. Your VOR/localizer receiver failed its last three 30-day checks, but you only plan to use GPS as primary navigation. Is the aircraft legal for IFR? **A:** Only if the GPS meets IFR navigation requirements on its own (WAAS-capable, or otherwise satisfying 91.205's navigation equipment requirement for the route) and any inoperative VOR equipment has been properly handled under 91.213 — placarded inoperative if not required for the flight, with a mechanic's determination. The VOR check itself (91.171) is only required if you intend to actually use VOR for IFR navigation.
14. What are the tolerances for a VOR operational check, and what methods are acceptable? **A:** $\pm 4^\circ$ using a VOT (VOR test facility) or a certified ground checkpoint, $\pm 6^\circ$ using a certified airborne checkpoint or a prominent ground landmark, or $\pm 4^\circ$ when comparing two VOR receivers against each other (dual system check). The date, place, bearing error, and your signature must be logged.
15. If your G5 (or similar STC'd glass instrument) is installed, does that mean you no longer need your old mechanical altimeter or airspeed indicator? **A:** Not necessarily — check the specific STC. Many glass-instrument STCs require retaining the original mechanical altimeter, airspeed indicator, and/or turn

coordinator because the new unit's pitot-static data isn't certified as primary, only supplementary. The 91.213 decision tree (check the POH/KOEL, then 91.205, then the type certificate data sheet, then any STCs) is what actually determines whether a "broken" old instrument grounds the aircraft — not just whether a modern replacement is present.

16. What anti-icing/de-icing equipment might be found on an IFR-capable airplane, and what should you know about its limitations? **A:** Pitot heat (nearly universal), and depending on the airplane, airframe boots or weeping-wing systems, propeller anti-ice, and heated windshield panels. None of these certify a normal single-engine trainer for flight into known icing — even an aircraft with some anti-ice equipment typically remains prohibited from known icing conditions unless specifically certified for it, and equipment limitations (e.g., boot cycle timing, maximum ice accumulation before effectiveness drops) should be understood before relying on them.

17. What's the purpose of an instrument flight deck check before an IFR flight, and what are you actually checking? **A:** Confirming the aircraft's instruments and IFR-required equipment are functioning correctly and the aircraft is airworthy for instrument flight — attitude indicator erect and stable, altimeter within tolerance of field elevation, DG aligned with the magnetic compass, turn coordinator/ball responding correctly to taxi turns, and navigation equipment tuned and identified.

Certificates, privileges, and currency. Now let's talk about you, not the airplane.

1. What are the eligibility requirements for an instrument rating? **A:** Hold at least a current private pilot certificate (or be concurrently applying for one) with the appropriate category/class rating; be able to read, speak, write, and understand English; pass the instrument knowledge test; and meet the aeronautical experience requirements of 61.65.

2. What's the aeronautical experience breakdown behind the instrument rating? **A:** 50 hours of cross-country PIC time (10 hours in airplanes), and 40 hours of actual or simulated instrument time in the required areas of operation, of which 15 hours must be received from an authorized instructor holding an instrument-airplane rating — including 3 hours of instrument training within 2 calendar months before the practical test, and one cross-country flight under IFR with a filed flight plan covering at least 250 nautical miles along airways or as routed by ATC, with an instrument approach at each airport and three different kinds of approaches using navigation systems.

3. What are the privileges and limitations of an instrument rating? **A:** It allows you to file and fly under IFR, operate in weather below basic VFR minimums (down to the applicable approach minimums), operate in Class A airspace, and — for a commercial pilot — carry passengers for compensation beyond 50 nautical miles or at night. It doesn't remove any category/class/type limitations already on your certificate.

4. What's required to remain current to act as PIC under IFR? **A:** Within the preceding 6 calendar months: 6 instrument approaches, holding procedures and tasks, and intercepting and tracking courses using navigational electronic systems. This can be done in actual or simulated instrument conditions, in the aircraft or an approved simulator/flight training device.

5. You haven't flown any IFR in 7 months. What are your options? **A:** You've lost currency but you're within the grace period — you have another 6 months (through the 12th month) to get current again by flying the 6 approaches/holding/intercept-tracking tasks with a safety pilot, without needing an instructor. If you go

beyond 12 months total without currency, you need an Instrument Proficiency Check (IPC) with a CFII to reinstate IFR privileges.

6. Who can act as your safety pilot, and what do they actually need to have? **A:** They need to hold at least a private pilot certificate with the appropriate category and class rating for the aircraft (not necessarily the ability to act as PIC of that specific aircraft — category/class/type is what matters, not full PIC eligibility), a current medical certificate, adequate vision forward and to the sides of the aircraft, and the aircraft must have functioning dual controls.

7. When can you log an instrument approach? **A:** In actual instrument conditions, you can log it as soon as you're in IMC at any point up to and including the final approach segment — you don't have to fly it all the way to minimums if you break out early. In simulated conditions (a view-limiting device), the approach must be flown down to minimums to be logged.

8. What is Part 68 BasicMed, and does it apply to instrument flying? **A:** An alternative to holding an FAA medical certificate for certain flights (aircraft under 6 occupants and 12,500 lbs, no more than 6 passengers, not for compensation or hire). It applies equally to IFR or VFR flight, as long as you stay at or below 18,000 ft MSL and 250 KIAS and meet BasicMed's other requirements.

Weather. Say you're planning today's flight for real. Let's start with the briefing.

18. How can you obtain a weather briefing, and what types are there? **A:** 1800wxbrief.com, the ForeFlight/Garmin Pilot app, AviationWeather.gov, or Flight Service by phone or radio. A Standard briefing is complete, used for the first briefing of the day or after 6+ hours since the last one. An Abbreviated briefing updates a previous one. An Outlook briefing is for planning when departure is 6+ hours away, and must be followed by a standard or abbreviated briefing closer to departure.

19. What are AIRMETs and SIGMETs, and how long is each valid? **A:** SIGMETs cover hazards significant to all aircraft — severe icing, severe/extreme turbulence, dust/sandstorms below 3 SM visibility, volcanic ash — valid up to 4 hours (6 for hurricanes). Convective SIGMETs cover severe thunderstorm activity, valid 2 hours. AIRMETs cover less severe conditions of particular concern to lighter aircraft, issued every 6 hours: Sierra (IFR/mountain obscuration), Tango (moderate turbulence, sustained winds ≥ 30 kt, low-level wind shear), Zulu (moderate icing and freezing levels).

20. What types of structural icing are there, and what conditions does icing require? **A:** Rime ice — rough, milky, forms quickly from small supercooled droplets, typical in stratiform clouds. Clear ice — smooth, hard, transparent, the most dangerous and hardest to remove, from larger supercooled droplets that spread before freezing. Mixed ice — a combination of both. Icing requires visible moisture and a temperature at or below freezing at the airframe surface.

21. What is a false glideslope, how do you avoid it, and how would you recognize one? **A:** A false glideslope indication can occur at certain angles well above the true glideslope, often around double the actual glidepath angle. It's avoided by intercepting the glideslope from below at the published glideslope intercept altitude, not from above — if ATC vectors you above the normal intercept altitude, be cautious about "catching" the needle before verifying your position against the final approach fix. A telltale sign is an unusually steep required descent rate to stay on the needle.

22. What's the difference between a localizer's and a VOR's service volume, and why does it matter? **A:** A localizer's usable service volume is roughly 18 nautical miles from the antenna and narrows as you get farther away — about 10° wide at 18 nm, widening to roughly 35° at 10 nm. Outside that volume, a "live" localizer needle isn't reliable, even if it appears to give a sensible indication — a common trap when you're well off course at long range and the needle looks plausible.

23. Flying from an area of very cold, high-pressure air to an area of hot, low-pressure air without resetting your altimeter — are you too high or too low, and what's the risk? **A:** You'd actually be lower than your altimeter indicates when flying toward lower pressure/warmer temperature without resetting — "high to low, look out below." (Conversely, flying from low pressure to high pressure without an update, the altimeter under-reads and you're higher than indicated.) On an approach, the practical risk is descending below a safe altitude or minimums while the altimeter still shows you comfortably above them.

Cross-country and IFR flight planning. Weather's checked. Now build the flight.

24. What does an IFR flight plan require, and how do you file, activate, and close one? **A:** Aircraft identification and type/equipment suffix, true airspeed, departure point, requested route and altitude, destination, and estimated time en route plus fuel on board, among the standard elements. File through Flight Service, ATC (a "pop-up" clearance, workload permitting), or a commercial service like ForeFlight — ideally at least 30 minutes before departure. An IFR flight plan in controlled airspace is automatically opened when you receive your clearance and automatically closed by ATC/tower on landing at a towered field; at a nontowered field, you must close it yourself with Flight Service or by radio.

25. What are the elements of a standard ATC clearance, and what's the mnemonic? **A:** CRAFT — Clearance limit, Route, Altitude, Frequency (departure), Transponder code. Example: "Cleared to [destination]. Radar vectors to [fix], then as filed. Maintain 2,000, expect 7,000 ten minutes after departure. Departure frequency 125.65. Squawk 4211."

26. What's an abbreviated departure clearance, and what does it still include? **A:** Used when your filed flight plan is approved with little or no revision — it still includes the clearance limit, any assigned departure procedure and transition, altitude, frequency, and transponder code, just abbreviated in phrasing (e.g., "cleared as filed").

27. What's the difference between a release time, a clearance void time, and a notify time for an IFR departure from a nontowered airport? **A:** The release time is the earliest you may depart. The clearance void time is the latest you may depart — if you haven't departed by then, the clearance is void and you must contact ATC. The notify time is the latest you should advise ATC if you haven't departed, if a separate notify time was given.

28. What's the difference between a SID and an ODP, and how do you tell them apart on a chart? **A:** An ODP (Obstacle Departure Procedure) exists purely for obstacle clearance and is designed to be flown without ATC involvement if not otherwise cleared; it's noted with "(OBSTACLE)" in the takeoff minimums section. A SID (Standard Instrument Departure) also provides obstacle clearance but is primarily designed to simplify clearance delivery and reduce ATC/pilot workload on a busy departure — it's assigned by ATC via a clearance.

29. What is a nonstandard takeoff minimum and climb gradient, and what determines whether you can legally use a given SID/ODP? **A:** Under Part 91, no takeoff minimums are legally required unless the departure procedure specifically mandates them, but a published departure procedure may require a minimum climb gradient (e.g., 260 ft/NM) to a specified altitude to guarantee obstacle clearance. Before accepting that departure, you need to know your airplane can actually achieve that climb gradient at the current weight, temperature, and wind — a climb gradient is a distance-based requirement, not directly the same as a feet-per-minute rate of climb, so it has to be converted using your expected groundspeed.
30. What do you check before departing on any IFR flight, per 91.103? **A:** NOTAMs, weather reports and forecasts, known ATC delays, runway lengths at airports of intended use, alternatives if the flight can't be completed as planned, and fuel requirements — takeoff and landing performance data for any flight.
31. What fuel is required to legally depart IFR? **A:** Enough to fly to the first point of intended landing, then to the alternate if one is required, then to fly for 45 minutes at normal cruise after that.
32. When is an alternate airport required, and what are the weather minimums for filing one (the "1-2-3" rule)? **A:** An alternate isn't required if, for at least one hour before through one hour after your estimated time of arrival, the forecast ceiling and visibility at the destination are at least 2,000 feet and 3 statute miles (for an airport with a published instrument approach). If the destination doesn't meet that, you must file an alternate that meets its own minimums: 600-2 (ceiling/visibility) if it has a precision approach, or 800-2 if only non-precision approaches are available — or the airport's own published alternate minimums if different.

Clearances, lost comm, and holding. You're airborne now. Let's talk about staying legal when things don't go as briefed.

33. If you don't fully understand an ATC clearance, or believe it might not be intended for you, what should you do? **A:** Ask for clarification or a full readback confirmation before complying — don't guess. A clearance inconsistent with your aircraft's performance or navigation capability, or one that might be intended for another aircraft with a similar call sign, is exactly the kind of thing worth double-checking rather than assuming is correct.
34. What is PIC emergency authority? **A:** In an in-flight emergency requiring immediate action, the pilot in command may deviate from any rule of Part 91 to the extent required to meet that emergency, though a written report may be requested by ATC afterward.
35. What altitude do you fly, and what route, if you lose two-way radio communication under IFR? **A:** Route: fly the route you were **A**ssigned in the last clearance; if being radar **V**ectored, fly direct to the fix/route/airway specified in the vector clearance; if none of those, fly the route you were told to **E**xpect in a further clearance; otherwise, fly the route you **F**iled (AVEF). Altitude: fly the highest of the altitude you were **A**ssigned, the altitude you were told to **E**xpect, or the **M**inimum en route altitude for the route segment (AIM's altitude mnemonic, sometimes remembered alongside AVEF).
36. If your clearance limit is a fix rather than the destination and you haven't received further clearance, what do you do at that fix? **A:** If you've received an Expect Further Clearance (EFC) time, hold at that fix until the EFC time, then proceed on your filed route. Only leave the clearance limit fix to begin an approach at your ETA (or EFC time, whichever is later) if the clearance limit is a fix from which an approach begins — otherwise, hold and wait for the EFC time before proceeding to a fix where an approach can be

commenced.

37. What are the standard holding pattern parameters unless otherwise instructed? **A:** Right-hand turns, one-minute legs (1.5 minutes above 14,000 ft). Begin timing the outbound leg when abeam the fix (or when the needle flips from TO to FROM), and adjust the outbound leg length as needed so the inbound leg is the correct duration.

38. Describe the three holding pattern entry procedures. **A:** Direct entry — cross the fix and turn directly into the holding pattern. Parallel entry — cross the fix, turn to parallel the inbound course outbound, then after about a minute turn through roughly 210° to intercept the inbound course. Teardrop entry — cross the fix, turn to a heading about 30° offset from the inbound course into the pattern, fly outbound about a minute, then turn about 210° to intercept the inbound course. Any entry that keeps you within the protected area is acceptable — there's no requirement to use the "textbook" entry for your approach angle.

39. What should you do with your wind correction angle when transitioning from the inbound to the outbound leg of a hold? **A:** Roughly triple the wind correction angle (in the opposite direction) on the outbound leg compared to what's needed to hold the inbound course — since you're only trying to correct for one leg's worth of drift on the way out to set up a clean inbound leg.

40. When must you comply with a charted procedure turn, hold-in-lieu-of-procedure-turn, or teardrop entry on an approach, and when can you skip it? **A:** It must be flown unless: you're given a straight-in approach clearance, you're being vectored to the final approach course by ATC, the approach is via a DME arc, "NoPT" is published for your route of arrival, or you're making a timed approach from a holding fix.

Navigation systems. Let's get into how you're actually finding your way to the airport.

41. How does VOR navigation work? **A:** A VOR station transmits a reference phase signal that's the same in all directions and a second, variable-phase signal whose phase relative to the reference changes depending on the radial you're on — the receiver compares the two signals' phase difference to determine your bearing from the station.

42. How does a localizer differ from a VOR, and why is it more accurate? **A:** A localizer transmits two overlapping signals at different modulation frequencies (90 Hz and 150 Hz) rather than a phase-comparison signal; on the extended runway centerline, equal amounts of each frequency are received, and being left or right of center shifts the balance toward one frequency or the other. This gives a much narrower, more precise course than a VOR's phase-comparison method — roughly a 700-foot-wide course at the runway threshold.

43. How does a glideslope work? **A:** The same frequency-overlap principle as a localizer, but arranged in the vertical plane from a separate antenna near the runway touchdown zone, giving vertical guidance down to the runway.

44. What's the difference between LNAV, LP, LPV, and LNAV/VNAV approaches? **A:** LNAV provides lateral (course) guidance only, using GPS, with a relatively wide final-segment CDI sensitivity (about 0.3 NM full-scale). LP (Localizer Performance) provides lateral guidance with localizer-like sensitivity (roughly $3-6^\circ$ full-scale, narrowing as you approach the runway) but no vertical guidance. LNAV/VNAV and LPV both add vertical guidance — LPV specifically uses WAAS to provide ILS-like precision approach guidance down to a

Decision Altitude, while LNAV/VNAV uses barometric or WAAS vertical guidance to a DA with typically higher minimums than LPV.

45. What is RNAV, broadly, and what are the different underlying technologies? **A:** Area navigation — the ability to fly a direct or defined path that isn't limited to flying directly to/from a single ground-based navaid. It can be implemented via GPS (most common in GA), VOR/DME, DME/DME, or inertial reference systems (still used in some airline/transport aircraft as a backup).

Instrument approach procedures. Now bring it down to the runway.

46. What's the difference between a precision and a non-precision approach? **A:** A precision approach (ILS, or GPS-based LPV) provides both lateral and vertical guidance down to a Decision Altitude (DA) — you either continue visually from the DA or execute a missed approach immediately, without leveling off at the DA. A non-precision approach provides lateral guidance only (or vertical guidance not accurate enough to be "precision"), descending to a Minimum Descent Altitude (MDA) which you may level off at and fly until reaching the missed approach point.

47. What is a Visual Descent Point (VDP), and how is it depicted? **A:** A point on a non-precision final approach, depicted as a black triangle on the approach chart's profile view, marking where a normal descent from the MDA to the runway can begin once the required visual references are in sight — descending before the VDP risks an excessively steep visual segment.

48. What conditions must be met to descend below the MDA or DA? **A:** The aircraft must be in a position to make a normal descent to landing using normal maneuvers; the flight visibility must not be below the minimum specified for the approach; and for Category I approaches, at least one of the specified visual references for the runway must be distinctly visible and identifiable (runway threshold, runway markings, runway lights, approach lighting system, threshold lights/markings, touchdown zone lights/markings, or a visual glideslope indicator). If the approach lighting system is the only reference in sight, descent may only continue to 100 feet above touchdown zone elevation unless the red terminating bars or red side-row bars (or another qualifying reference) also become visible.

49. What are aircraft approach categories, and what determines which one you fly? **A:** Categories A through E (and beyond) are based on the aircraft's approach speed — Category A is under 91 knots, B is 91–120 knots, C is 121–140, D is 141–165. Your category (and thus your minimums) is based on 1.3 times your stall speed in the landing configuration at maximum certificated landing weight, not simply the approach speed you choose to fly, and you may not use a category slower than the one your aircraft's actual speed requires.

50. What is a circling approach, and what's the protected maneuvering area based on? **A:** A circling approach descends to a circling MDA, and once the runway environment is in sight, the aircraft maneuvers visually to land on a runway not aligned with the approach course, remaining at or above the circling MDA until a normal descent to land can be made. The protected area is an arc-defined radius from each runway threshold, sized according to the aircraft's approach category (and whether standard or expanded circling minimums are in use) — losing sight of the airport or leaving that protected area requires an immediate missed approach.

51. What must you do immediately upon reaching a missed approach point (or DA, for a precision approach) without the required visual references? **A:** Execute the missed approach procedure promptly — for a precision approach, this means immediately, without leveling off at the DA first. Apply appropriate power, establish the correct pitch attitude, configure the aircraft per the manufacturer's procedure, and follow the published (or ATC-assigned alternate) missed approach procedure, advising ATC if unable to comply with any part of it.
52. What is a Terminal Arrival Area (TAA), and what's its purpose? **A:** A "T"-shaped design used mostly on RNAV approaches, with two Initial Approach Fixes and a dual-purpose Intermediate Fix/IAF that includes a hold-in-lieu-of-procedure-turn, feeding into the final approach fix. It provides a transition from en route to the approach with minimal ATC/pilot interaction and its own published minimum altitudes (replacing the usual minimum safe altitude circle), typically allowing clearance much farther from the airport than a conventional approach.
53. What's the difference between a visual approach and a contact approach? **A:** A visual approach is ATC-authorized once the pilot has the airport (or the aircraft to follow) in sight, requires a ceiling of at least 1,000 ft and 3 SM visibility at the destination, and clears the pilot to proceed visually clear of clouds — normal VFR cloud clearances don't apply since it's still an IFR procedure. A contact approach requires only 1 SM flight visibility and clear-of-clouds operation, must be requested by the pilot (ATC can't initiate it), and requires the destination to have a published instrument approach — it lets the pilot proceed to land without flying the full published procedure.

Flight instruments and instrument errors. Let's step back to the panel itself.

54. How does the pitot-static system work, and what happens to the altimeter and airspeed indicator if the static port is blocked? **A:** The pitot tube senses ram air pressure; the static port senses ambient (still) air pressure; the airspeed indicator compares the two, while the altimeter and VSI use static pressure alone against a calibrated reference. If the static port is blocked, the altimeter and VSI will freeze at the pressure/altitude that existed when the blockage occurred, regardless of actual altitude change, and the airspeed indicator will become inaccurate — reading progressively lower than actual as you climb (since ram air pressure drops with altitude but the trapped static reference doesn't fall with it), and progressively higher than actual as you descend.
55. What are the gyroscopic instruments, and what two properties of gyroscopes do they rely on? **A:** The attitude indicator, heading indicator (directional gyro), and turn coordinator. They rely on rigidity in space (a spinning gyro resists changes to the plane of its rotation) and precession (a force applied to a spinning gyro produces a resulting force 90° later in the direction of rotation).
56. What magnetic compass errors should you know, and how do you remember them? **A:** Variation is the fixed difference between true and magnetic north at a location ("east is least, west is best" — subtract east variation, add west variation, going from true to magnetic). Deviation is compass-card error from the aircraft's own electrical/magnetic fields, unique to each airplane. Dip errors occur during maneuvering: on northerly headings, the compass initially shows a turn in the opposite direction during a turn and lags; on southerly headings it leads ("UNOS" — undershoot north, overshoot south, referring to when to stop your turn relative to the desired heading); during acceleration/deceleration on east/west headings, the compass shows a false turn indication ("ANDS" — accelerate north, decelerate south).

57. What's the primary/supporting instrument method of cross-check, and give an example for straight-and-level flight. **A:** Each flight parameter (pitch, bank, power) has one "primary" instrument that gives the most direct, immediate indication of that parameter in a given maneuver, with other instruments "supporting" that reading. In straight-and-level flight: pitch primary is the altimeter (supported by attitude indicator and VSI); bank primary is the heading indicator (supported by attitude indicator and turn coordinator); power primary is airspeed (supported by tachometer/manifold pressure).

58. What are the components of a glass cockpit — AHRS, ADC, PFD, MFD, and Flight Director — and what does each replace? **A:** The AHRS (Attitude and Heading Reference System) provides attitude and heading data, replacing the mechanical attitude indicator, directional gyro, and turn coordinator. The ADC (Air Data Computer), fed by the pitot-static system, replaces the airspeed indicator, altimeter, and VSI. The PFD (Primary Flight Display) graphically presents this "big six" information on one screen; the MFD (Multi-Function Display) shows moving maps, weather, traffic, and system data, and can typically revert to display PFD information if the primary PFD fails. A Flight Director computes and displays a desired flight path/pitch-bank cue to assist hand-flying or drive the autopilot.

Spatial disorientation and unusual attitudes. Now let's talk about when the panel and your body disagree.

59. What is spatial disorientation, and why is it especially dangerous under instrument conditions? **A:** A false perception of the aircraft's attitude, position, or motion, caused by relying on the body's vestibular and proprioceptive senses rather than the instruments — those senses are unreliable in flight, especially without outside visual reference (IMC or night). The remedy is trusting and cross-checking the flight instruments rather than physical sensation, and maintaining a disciplined instrument scan.

60. How do you recognize and recover from a nose-low unusual attitude by reference to instruments? **A:** Recognize increasing airspeed and a nose-low, often increasing-bank attitude indication. Reduce power, level the wings using coordinated aileron and rudder, then smoothly raise the nose to level flight — correcting bank and pitch essentially together rather than one fully before the other.

61. How do you recognize and recover from a nose-high unusual attitude by reference to instruments? **A:** Recognize decreasing airspeed (and possibly an approaching stall warning) with a nose-high attitude. Apply full power, level the wings, and smoothly lower the nose to return to level flight.

62. What should you do if you inadvertently enter IMC while not instrument-rated, or otherwise experience unintended IMC? **A:** Transition immediately and fully to the instruments, hold wings level and a constant airspeed, avoid abrupt control inputs, and either turn back toward known VMC (a standard-rate 180°) or, if instrument-rated and IFR-capable, contact ATC and proceed accordingly. Avoid the temptation to look outside for visual references once committed to the instruments — that's exactly what triggers spatial disorientation in this scenario.

Emergency operations. Let's stack a couple of failures together and see how you handle it.

63. What are the general procedures for lost communications, beyond route and altitude? **A:** First, try to reestablish communication — check the frequency, volume, headset connection, and try a different frequency or a nearby facility. Squawk 7600. If unable to reestablish contact, proceed via the AVEF route

and MEA-based altitude logic, and begin your approach at your filed/estimated time of arrival (or as close to it as operationally possible) if no other guidance is available.

64. If your primary flight instruments (attitude indicator, etc.) fail or become unreliable during an approach, what's your plan? **A:** Recognize the failure or inaccurate indication, advise ATC, and rely on your secondary/backup instruments and cross-check to continue safe control of the aircraft — most modern trainers retain a standby attitude source or can fly a non-precision approach using the remaining reliable instruments (airspeed, altimeter, heading, turn coordinator) even without the primary attitude display.

65. What is your responsibility, and ATC's role, if you declare an emergency? **A:** As PIC, you have the authority to deviate from any regulation to the extent required to meet the emergency, though you may be asked to file a report afterward. ATC can provide priority handling and vectors — declaring early, rather than waiting until the situation is critical, gives everyone more room to help.

Postflight. Last one. Bring it home.

66. What must be documented after an IFR flight? **A:** Any in-flight or postflight discrepancies noted with the aircraft's instruments or equipment, servicing requirements, and flight time/approaches logged for currency purposes. The PIC remains responsible for ensuring any discrepancy affecting airworthiness is addressed before the aircraft's next flight.